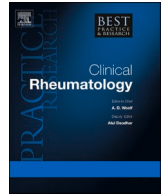




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Utility of polygenic risk scores to aid in the diagnosis of rheumatic diseases

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ABSTRACT

Rheumatic diseases (RDs) are characterized by autoimmunity and autoinflammation and are recognized as complex due to the interplay of multiple genetic, environmental, and lifestyle factors in their pathogenesis. The rapid advancement of genome-wide association studies (GWASs) has enabled the identification of numerous single nucleotide polymorphisms (SNPs) associated with RD susceptibility. Based on these SNPs, polygenic risk scores (PRSs) have emerged as promising tools for quantifying genetic risk in this disease group. This chapter reviews the current status of PRSs in assessing the risk of RDs and discusses their potential to improve the accuracy of the diagnosis of these complex diseases through their ability to discriminate among different RDs. PRSs demonstrate a high discriminatory capacity for various RDs and show potential clinical utility. As GWASs continue to evolve, PRSs are expected to enable more precise risk stratification by integrating genetic, environmental, and lifestyle factors, thereby refining individual risk predictions and advancing disease management strategies.

1. Introduction

Rheumatic diseases (RDs) represent a diverse and complex spectrum of both autoimmune and autoinflammatory diseases that have a significant impact on global health. The burden of these conditions is substantial; the prevalence of autoimmune diseases within the general population is estimated to be between 5% and 10%, a figure that has steadily increased over recent decades. This trend highlights the potential influence of environmental factors, such as dietary changes and climate variations, on disease prevalence [1]. Additionally, the progressive disability associated with RDs not only diminishes work-life quality but also escalates socioeconomic

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costs [2,3].

Rheumatology covers a broad discipline that includes more than 100 different RDs, each characterized by a unique set of clinical symptoms and complex pathology. In addition to their primary impact on joints, these diseases often have extra-articular manifestations that affect the eyes, skin, lungs, kidney, cardiovascular system, and gastrointestinal system.

Most RDs are complex or multifactorial entities because their pathogenesis involves a combination of multiple genetic factors and a diverse array of environmental and lifestyle effects. In this chapter, we explore the complex interplay of human genetics in the pathogenesis of RDs and its significant influence on their clinical variability. Unlike clinical risk factors, which typically manifest later in life, a genetic predisposition is established at conception and remains constant throughout an individual's lifespan. This inherent genetic risk allows clinicians to use genetic data to predict long-term disease risk well before clinical symptoms appear, potentially enabling more effective lifestyle modifications.

First, we will discuss existing biomarkers and their limitations in diagnosing RDs. We then review how polygenic risk scores (PRSs), derived from the aggregated genetic risk from common alleles identified through genome-wide association studies (GWASs), have emerged as a crucial method for quantifying the risk attributable to genetics. Although numerous PRSs have been developed for various diseases and they show significant associations with disease risk, their clinical utility, particularly in the realm of RDs, is still being explored. Therefore, the aim of this chapter is to review the status of PRSs in regard to the risk of RDs and to discuss the potential of PRSs to improve the diagnosis of these complex diseases.

2. Limitations of biomarkers in rheumatic diseases

Inflammatory arthritis conditions, such as rheumatoid arthritis (RA) and spondyloarthritis (psoriatic arthritis (PsA) and axial spondyloarthritis (ax-SpA)) are the most prevalent RDs worldwide, while other rheumatic conditions, such as systemic lupus erythematosus (SLE) and scleroderma (SSc), are less frequently diagnosed [4]. Classification criteria for RDs have been developed over the years for better patient assortment and for providing personalized medicine with individually targeted treatments. These classification criteria often include biomarkers, including autoantibodies such as rheumatoid factor (RF) and anti-cyclic citrullinated peptides (*anti*-CCP) in RA [5] or anti-nuclear antibodies (ANA) in SLE [6]; genetic alleles such as HLA-B27 in ax-SpA [7]; laboratory tests such as erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) in RA [5]; and C3/C4 in SLE [6]. Despite the utility of these biomarkers, which lead clinicians through patient classification for homogenizing RDs and selecting standard treatments, their predictive values vary widely, and their diagnostic limitations are not well established.

2.1. Rheumatoid arthritis

RA is a chronic autoimmune disease that affects approximately 1% of the population worldwide [8] and is characterized by inflammation of the synovial joints, which leads to pain and stiffness and, in many cases, irreversible bone damage. The continuous inflammation in these patients can also affect extra-articular organs such as the heart, kidney, lung, eye, skin, and digestive and nervous system [9]. The pathogenic mechanisms responsible for RA onset are not clear, and the course of the disease can fluctuate between sporadic flares-up to ongoing disease, leading to severe joint damage, which reduces the quality of life and life expectancy [10].

Rheumatoid factor (RF) is an autoantibody against the Fc region of immunoglobulin G (IgG) that is found in most RA patients (80%) and is associated with more severe disease and greater impairment; hence, its role surpasses the diagnostic aim (66% sensitivity, 90% specificity), and RF is also considered a prognostic biomarker. Its diagnostic specificity is restricted due to its presence in healthy individuals and individuals with other inflammatory conditions (malignancies, infectious diseases and other RDs), but the coexistence of high titres of RF and other autoantibodies (*anti*-CCP) increases its post-test probability for an RA diagnosis [11–13]. *Anti*-CCP (e.g., *anti*-CCP2 and *anti*-CCP3) is a group of autoantibodies against citrullinated proteins that is found in two-thirds of RA patients. Its diagnosis prediction is higher than the one provided by RF, being related with a poor prognosis with greater disability and more erosive disease. It has higher sensitivity and specificity than RF (70% and 96%, respectively). Its limitations lie in the presence of *anti*-CCP in healthy individuals (up to 5%) and other diseases (e.g., tuberculosis and other RDs) [11–13].

A global limitation for all autoantibodies is the presence of seronegative RA, wherein autoantibodies are absent. This condition is present in 20–30% of patients and becomes the first limitation for autoantibody employment in RA diagnosis [14]. Moreover, autoantibodies are not useful as monitoring biomarkers due to their fluctuations during the disease course. However, several research papers have reported the utility of autoantibodies for assessing likelihood of treatment response, especially for patients receiving biologic disease-modifying antirheumatic drugs (bDMARDs). In fact, RF has been found to increase the probability of a successful response to B-cell depletion therapy, and high *anti*-CCP titres have been related to a better clinical response to a few bDMARDs, such as rituximab, abatacept or anti-tumour necrosis factor (*anti*-TNF) [11–13]. These results have not been extrapolated to all RA populations, and a lack of consistency between studies has led to the relegation of autoantibodies to a secondary role as treatment biomarkers.

The erythrocyte sedimentation rate (ESR) measures the speed at which erythrocytes settle into the bottom of a vertical tube. The ESR is not a specific biomarker of RA inflammation, but it is closely related to fibrinogen (the inflammation acute-phase reactant (APR)). C-reactive protein (CRP) is a pentraxin that is considered an APR, and it is produced predominantly by hepatocytes after interleukin-6 (IL-6) stimulation. In addition to APR, it is considered an immune regulator with multiple functions, such as complement activation, neutrophil recruitment through IL-18 or the production of matrix metalloproteinases. The advantages and limitations of their use in treating RA are summarized in Table 1.

Multibiomarker disease activity (MBDA) is a score that focuses on the detection of 12 biomarkers: CRP, leptin, resistin, serum

amyloid (SAA), matrix metalloproteinase-1 (MMP-1), MMP-3, vascular cell adhesion molecule-1 (VCAM-A), epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), IL-6, TNF receptor type 1 (TNF-R1) and human cartilage glycoprotein 39 (YKL-40). It was developed for monitoring RA, but controversial results have been published concerning its correlation with clinical activity scores. Its usage is not cost-effective in daily clinical practice [12,15].

Anti-carbamylated proteins (anti-CarP) have been identified in RA patients, but their sensitivity and specificity for RA diagnosis do not surpass those of RF and *anti*-CCP. 14-3-3eta is a chaperonin explored in RA diagnosis that reclassifies some patients previously classified as “seronegative RA”. A different *anti*-CCP antibody, anti-mutated citrullinated vimentin (*anti*-MCV), has been explored for improving RA diagnosis, but its specificity is similar to that of *anti*-CCP2, limiting its implementation in clinical practice. All of these new biomarkers have been identified in several research papers, but their routine implementation in clinical practice has not been achieved, a major limitation [12,13].

2.2. Spondyloarthropathies

AxSpA encompasses a heterogeneous group of chronic inflammatory arthropathies primarily affecting the spine and the sacroiliac joints, although peripheral symptoms and extra-articular manifestations are also common. Both radiographic and nonradiographic forms of axSpA have been described. The radiographic form, known as ankylosing spondylitis (AS), traditionally characterizes the more severe, classic presentation with evident changes on radiographs [16]. In contrast, nonradiographic axSpA (nr-axSpA) describes early or milder cases where radiographic evidence is not yet apparent, although both types share the same underlying inflammatory mechanisms [17]. The prevalence of axSpA is estimated to be between 1.2% and 1.8%, with a significant predominance in individuals under 40 years of age [18]. This presents a considerable public health challenge due to its substantial impact on the public health care system, primarily because it affects individuals in their prime working years, potentially leading to chronic health issues and increased health care costs.

AxSpA is a highly heritable rheumatic condition in which 25% of this heritability is due to HLA-B27 (human leukocyte antigen B27); this heritability is one of the strongest genetic associations found in a polygenic disease (odds ratio near 60) [19]. HLA-B27 is one of the most commonly used genetic markers across diseases; it is strongly associated with axSpA (nearly 90% in individuals with established disease and less common in individuals with nr-axSpA (70%)) and is a core biomarker in the diagnosis of spondyloarthritis [20,21], forming an essential part of the ASAS classification criteria for spondyloarthritis [7]. It has high sensitivity and low specificity due to its prevalence in the healthy population, which is up to 25%, whereas only 1–6% of HLA-B27-positive individuals develop axSpA. This finding supports the probable contribution of other genes to the susceptibility to axSpA. Moreover, HLA-B27 is related to long-term radiographic progression in axSpA patients. It encompasses peripheral involvement, familial aggregation, and the presence of uveitis in its prognostic value [22].

Antibodies against CD74 (gamma chain of MHC class II implicated in inflammation, antigen presentation and B-cell differentiation) or CLIP (class-II associated invariant chain peptide, an extracellular portion of CD74) have been identified in axSpA patients (up to 85% compared to the healthy population (<15%)) and are related to early disease (their reactivity declines after the first year) [20]. They have higher sensitivity (85%) than HLA-B27 for an axSpA diagnosis (77%); therefore, they have been recently identified as newer

Table 1
Attributes of acute phase reactants in rheumatic diseases.

APR		Rheumatic Diseases		
		RA	axSpA	PsA
ESR	Benefit	- ACR/EULAR 2010 classification criteria. - Disease activity index (DAS28).	- Radiographic progression and osteoporosis.	- Correlated with joint tenderness and swelling, radiographic damage and synovitis in MUS.
	Drawback	- 40% RA normal values.	- 50% axSpA normal values. - Less reliable than CRP.	- 50% PsA normal values.
CRP	Benefit	- ↑ Healthy subjects (woman, elderly) and other diseases (infections, malignancies, obesity). - ACR/EULAR 2010 classification criteria. - Disease activity Indices (DAS28, SDAI). - Key role in bone destruction (RANKL). - Monitoring (disease activity, impairment, radiological damage, comorbidities). - Poor response to csDMARD.	- Spondyloarthritis feature in ASAS classification criteria. - Monitoring (correlation with BASDAI and ASDAS, CV disease and inflammation in MRI). - ↑ baseline is related with better treatment response (especially for <i>anti</i> -TNF). - ↑ r-axSpA instead of nr-axSpA.	- Correlated with joint involvement and synovitis in MUS.
	Drawback	- 40% RA normal values. - TCZ normalizes levels. - Levels modified by sex or age. - ↑ Other diseases (infections, malignancies, obesity, diabetes).	- 50% axSpA normal values.	- 50% PsA normal values.

Abbreviations: APR, acute phase reactants; ESR, erythrocyte sedimentation rate; CRP, C-reactive protein; RA, rheumatoid arthritis; axSpA, axial spondyloarthritis; PsA, psoriatic arthritis; RANKL, receptor activator of nuclear factor- κ B ligand; csDMARD, classical synthetic disease modifying antirheumatic drug; TCZ, tocilizumab; ASAS, Assessment of Spondyloarthritis International Society; BASDAI, Bath Ankylosing Spondylitis Disease Activity Index; ASDAS, Ankylosing Spondylitis Disease Activity Score; CV, cardiovascular; MRI, magnetic resonance imaging; TNF, tumour necrosis factor; r, radiographic; nr, nonradiographic; MUS, musculoskeletal ultrasonography.

diagnostic biomarkers, but inconsistent results have been published concerning them. Inconsistencies between studies and populations assessed complicate its total application in clinical practice [23–25].

Antibodies against protein phosphatase magnesium-dependent 1A (PPM1A) have also been identified in axSpA patients and are related to radiographic changes and the response to TNF inhibitors, but their sensitivity (66%) and specificity (73%) are insufficient for diagnostic purposes [23]. Other antibodies, such as *anti*-MCV (14%), anti-mycobacterial HSP 65 kDa (HSP65) and anti-flagellin (CBir-1), have been identified in axSpA patients, but no correlation with predominant disease (axial or peripheral) or disease activity has been found; hence, their role as biomarkers is uncertain [20].

The APRs are related to monitoring rather than diagnosing, and their main attributes are summarized in Table 1. They have low sensitivity and specificity as diagnostic biomarkers.

Several microRNAs (miRNAs, small noncoding RNAs that regulate posttranscriptional changes, mostly decreasing expression) have been identified as biomarkers in axSpA due to their high expression (concerning inflammatory genes), but their accuracy ranges from 0.6 to 0.9 (area under the curve, AUC), suggesting their inconsistency for axSpA diagnosis [23,24].

The IL-17/IL-23 pathway has become an important feature in axSpA pathogenesis. The levels of these cytokines have been shown to be increased in patients with axSpA and they decrease in responders under treatment with TNF inhibitors; for this reason, these cytokines have emerged as therapeutic targets. However, their relationship with active inflammation or disease activity has not been clearly defined. Other cytokines, such as TNF α or IL-6, are increased in axSpA patients (mainly early in the disease course), but their use as biomarkers for targeting or monitoring (e.g., disease activity, treatment response, or radiographic progression) is currently limited to research [20,23,24,26].

Higher urinary levels of C-terminal telopeptide of type II collagen (CTX-II) have been found in axSpA patients than in healthy individuals, and this finding correlates with inflammation on MRI and with radiographic progression. Furthermore, its levels decrease after TNF inhibitor treatment. This molecule is a biomarker of cartilage remodelling, as are YKL-40 (serum human cartilage glycoprotein-39) and COMP (cartilage oligomeric matrix protein) [20,23,24,26]. Other biomarkers of bone remodelling are DKK-1 (Dickkopf-1), sclerostin (both related to Wnt signalling) and cathepsin K, which are related to radiographic progression and bone formation, although inconsistencies have been identified among published studies. Finally, other biomarkers, such as MMP-3 (matrix metalloproteinase 3), VEGF (vascular endothelial growth factor), SAA (serum amyloid A, in combination with CRP improves early axSpA diagnosis and predicts the response to TNF inhibitors, but high costs and low availability limit its use), calprotectin (calcium and zinc binding protein located mainly in neutrophils, associated with disease activity, radiographic progression, treatment response and atherosclerosis) and several adipokines (visfatin, associated with radiographic progression and disease activity; and leptin, linked with radiographic progression), are increased in axSpA patients. The use of these molecules in clinical practice is uncertain, and they do not lead to higher sensitivity or specificity rates for diagnosing or monitoring axSpA [20,23,24,26].

The association of HLA-B27 with PsA is less weaker than that with axSpA and more important than its link with psoriasis (up to 35% of PsA patients) [27]. Moreover, it is more frequent when axial involvement is present in the disease and is related to radiographic progression, extraarticular involvement and a poor prognosis. Although HLA-B27 is considered one of the strongest genetic biomarkers for PsA, it carries a small fraction of the overall genetic risk [27].

The absence of RF is part of the Classification for Psoriatic Arthritis (CASPAR) criteria for PsA, as its presence could help to diagnose other inflammatory arthritis other than PsA, such as RA [28]. RF and *anti*-CCP are absent in nearly 90% of PsA patients [29]. The current sparse evidence about autoantibodies in PsA diagnosis leads them to a second level in diagnosis resources. Antibodies against nebullin-related anchoring protein (*anti*-NRAP) have been identified in PsA patients (83%) but not in RA patients (11%) or psoriasis patients (3%). Anti-A disintegrin and metalloproteinase with thrombospondin motif 5 (ADAMTS5) and *anti*-cathelicidin LL37 (LL37) have been employed to differentiate psoriasis from PsA, with accuracies of 0.84 and 0.87 (AUC), respectively. These promising autoantibodies for PsA diagnosis have not been validated in other cohorts, so their applicability is still unknown [29]. APR shares similar attributes to those mentioned above, and they are summarized in Table 1.

Cartilage and bone turnover biomarkers have been described as diagnostic biomarkers [29]. Greater COMP levels have been detected in PsA patients than in psoriasis or osteoarthritis patients, as well as in synovial fluid, but not in RA patients. Its levels were correlated with APR and disease activity. Its odds ratio is 1.001; therefore, its accuracy for PsA diagnosis is limited.

MMP-3 has been described as a diagnostic biomarker for PsA, but discordant studies have been published, and its accuracy for discriminating PsA from other conditions, such as psoriasis or healthy subjects, is moderate (AUC 0.7). Additionally, decreased TNF- α levels have been found in PsA patients who are responders to TNF inhibitors and they correlate with a better response to TNF inhibitor treatment. Studies concerning RANK-L, osteoprotegerin and DKK-1 have shown contradictory results and low accuracy in distinguishing PsA from other states, such as psoriasis or a healthy population (AUC 0.6). Moreover, RANK-L levels correlate with radiographic progression and disease severity [30,31].

Circulating miRNAs have been reported to be stable in PsA patients compared to healthy subjects; they seem stable, and their expression between patients is consistent. MiRNAs have been described in other inflammatory, malignant, and cardiovascular diseases. This research is still ongoing, and stronger evidence is needed before its implementation as a PsA biomarker [30].

Other serum biomarkers are C-X-C motif chemokine ligand 10 (CXCL10), whose levels are greater in psoriatic patients who develop PsA and decline after the first year of PsA diagnosis; calprotectin, which has good sensitivity (75%) and specificity (93%) for identifying PsA patients, has been reported increased in patients suffering from other types of inflammatory arthritis and, even, in atherosclerosis; interleukins such as IL-2 and IL-10, whose levels have been found to be elevated in PsA patients compared to those in psoriatic patients or healthy controls; IL-6, which has been found to be increased but with low specificity; and programmed death-1 (PD1), whose levels have been inversely correlated with disease activity scores, especially with the number of joints involved [29,30].

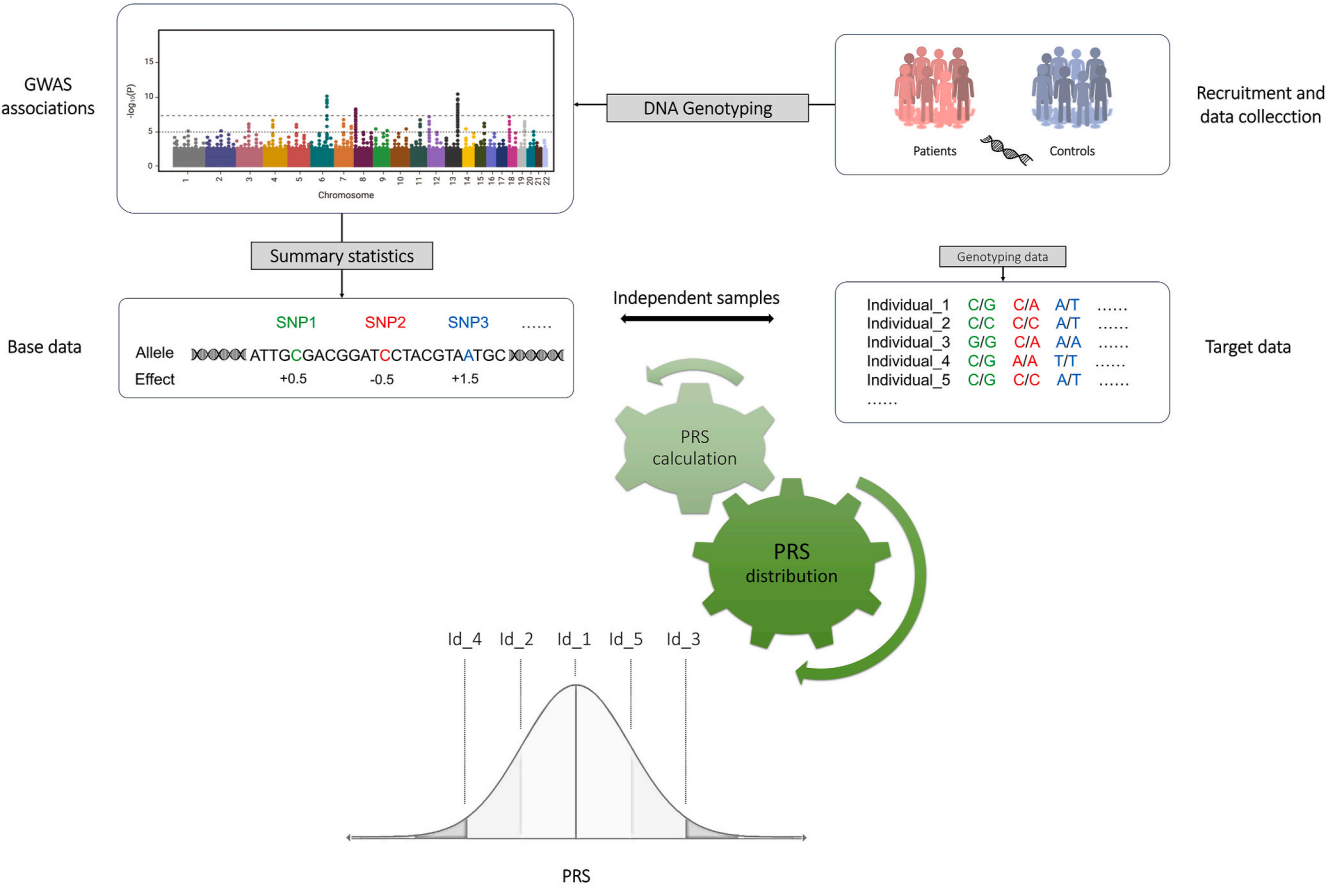


Fig. 1. Design of genome-wide association studies and development of a polygenic risk score. The process begins with the collection of genotypic data, either from specific study cohorts or through existing biobanks and repositories, utilizing mainly microarrays for common variants or more comprehensive methods such as next genome sequencing. These data undergo genetic association tests to identify the effect of each SNP on the target phenotype, accounting for confounders such as population stratification and controlling for multiple testing biases. Summary statistics from GWAS (base data) provide insights into the SNP effects, which are then used to reference genotype data from individual participants (target data). The PRSs are computed by aggregating the effect sizes of all risk alleles present in each individual. Abbreviations: GWAS, genome-wide association study; PRS, polygenic risk score; SNP, single-nucleotide polymorphism.

2.3. Other rheumatic diseases

There are several biomarkers used for the diagnosis of SLE [32,33]. Antinuclear antibodies (ANAs) are the main biomarkers needed for SLE classification according to the updated American College of Rheumatology/European Alliance of Associations for Rheumatology (ACR/EULAR) 2019 criteria, even though they are not specific for SLE (they are present in many autoimmune diseases such as Sjögren syndrome and in healthy individuals (up to 20%, especially in elderly individuals)) but are highly characteristic of SLE, which results in an elevated sensitivity (90%). ANA detection varies between techniques, with Hep-2 cells being the most common detection method used. SLE ANA negativity (6%) and the use of other ANA techniques (ELISAs) are sources of inaccuracies. Serum complement 3 (C3) and C4 are two of the biomarkers most commonly used for SLE monitoring: low levels indicate a possible flare (mainly renal or haematological). Both are used in the classification criteria for SLE, and when combined with ANA testing, the specificity for SLE diagnosis increases (>94%). *Anti-dsDNA* (double stranded deoxyribonucleic acid) antibodies are used for diagnosis (ACR/EULAR classification criteria) and monitoring, and higher levels can predict a flare (especially lupus nephritis (LN)). These methods have high specificity (96%) and low sensitivity (70%) for SLE diagnosis due to their fluctuations during the disease course. *Anti-Sm* (Smith) antibodies are common in SLE patients, with a specificity of 99% but a lower sensitivity (30%). These antibodies have not been detected in healthy individuals or individuals with other rheumatic diseases and are used for SLE classification. These factors are related to LN and worse outcomes. Proteinuria is a classic biomarker of LN in patients with SLE, but it can be detected in patients with many glomerular diseases. Several biomarkers of neuropsychiatric SLE, such as antiphospholipid antibodies (lupus anticoagulant, anticardiolipin, and anti-beta2-glycoprotein I), *anti-N-methyl-D-aspartate receptor (anti-NMDAR)* or anti-ribosomal proteins (*anti-RibP*), are elevated in patients with neuropsychiatric involvement. To date, although it is known that each of these diseases exhibit a specific antibody pattern that could be used as diagnosis and/or prognosis tools, no antibody or cytokine biomarker has been reported as a totally specific diagnosis biomarker for any rheumatic disease. Thus, the future genetic studies should be focus on the uncovering of specific genetic patterns.

3. Polygenic risk scores

Over the past fifteen years, our understanding of genetic susceptibility to complex diseases has significantly improved, primarily due to the increasingly powerful statistical capacity of GWASs. These studies are based on analysing the effects of genetic markers such as SNPs [34]. SNPs predominantly consist of single base substitutions within the DNA sequence and are estimated to represent 90% of the genetic variability in the human genome. In each GWAS, the frequency of the alleles that make up each SNPs analysed in a large series of individuals affected with the phenotype object of study is compared with those in a population of matched healthy individuals (Fig. 1). These studies typically involve the genotyping of hundreds of thousands to millions of SNPs, which are evenly distributed across the genome. This extensive coverage ensures a high likelihood of identifying associations with genetic markers. Technical advances in high-throughput genotyping platforms have made GWASs increasingly accessible to the scientific community. Extensive GWASs of a great number of diseases and traits have been conducted. Most studies have demonstrated the presence of polygenic influences on disease susceptibility [35].

Utilizing patterns of linkage disequilibrium (LD) among common genetic variants, arrays containing 500,000 to 1,000,000 ‘tag SNPs’ can be designed to capture a large proportion of common SNPs (~10 million). This approach enables the identification of trait-associated SNPs as representative of true causal variants, which are likely situated within the same block of trait-associated SNPs. One notable application of GWAS is the imputation of human leukocyte antigen (HLA) alleles [36]. The HLA region, also known as the major histocompatibility complex (MHC), is highly polymorphic and plays a critical role in the immune system by presenting pathogen-derived peptides to T cells [37]. GWASs facilitate the prediction of HLA alleles based on SNP data [38,39], which is crucial for studying diseases with a strong immune component.

The execution of larger GWAS is essential for the characterization of complex diseases, through the use of advancing strategies that allow to uncover the genetic architecture of their pathogenesis. The benefits of this will be greater as larger sample sizes become available and related and specific phenotypes are analysed [35]. These results will be used in different ways, such as drug repositioning, where potentially new pharmacological targets can be identified [40]. Drug repositioning based on GWAS studies is already being evaluated in preclinical and early-phase trials and could be expedited [41]. Other uses of these findings include Mendelian randomization, which in the context of complex diseases can be a fundamental tool for distinguishing causal from noncausal risk factors and prioritizing modifications of the former [42]; validation of functional hypotheses through the use of experimental techniques such as genome editing and/or the use of cellular models [43]; and the integration of GWAS findings with other omics data also offers opportunities for deeper insights (functional genomics) to understand how DNA variations lead to changes in gene expression or biological function, thereby influencing disease [43] and the development and utilization of PRSs.

PRS are estimates of an individual’s genetic susceptibility to a disease, calculated by summing the effects of each individual SNP marker (risk allele) across their genome. The contribution of each allele is weighted by its associated effect size, which is derived from the GWAS summary data [44] (Fig. 1). The PRS allows the integration of extensive genetic information to estimate disease susceptibility at the individual level, significantly improving our understanding of the genetic architecture of complex diseases and the ability to construct models of risk stratification of patients for some common diseases [45]. Thus, PRSs are potentially useful for the early diagnosis or prediction of the likelihood of developing complex diseases and could aid in designing early intervention strategies.

The methodologies for calculating PRS vary but generally involve selecting SNPs based on their association with the trait of interest. The simplest and most common method is the threshold-based PRS, where SNPs are included in the score if their association *p* values fall below a predetermined threshold. This method, however, might miss SNPs that individually have a small effect but

cumulatively could contribute significantly to the trait. Due to advances in statistical genetics, there are more sophisticated methods that utilize penalized regression (Lassosum) [46] and Bayesian approaches, such as Bayesian sparse linear mixed models [47], AnnoPred [48], PleioPred [49], PRS-CS [50], the algorithm by Newcombe et al. [51], LDpred [52] and LDpred-funct [53]. These methods aim to provide more accurate predictions by accounting for the uncertainty in SNP effects and the genetic correlation induced by LD. Overall, PRSs are powerful tools that potentially offer insights into genetic predispositions to diseases and enhance the capabilities of personalized medicine. The accuracy of these methods depends significantly on the quality of the underlying GWAS data and the appropriateness of the statistical models used for their calculation.

4. Polygenic risk scores in rheumatology

PRSs are emerging as pivotal tools in the field of personalized medicine and promise to revolutionize the management of many complex diseases, including RDs. With most rheumatic conditions characterized by moderate to high heritability yet a low prevalence [54], the application of PRSs holds significant potential for improving disease outcomes, both improving diagnostic precision and designing early intervention strategies. Basically, the ability of PRS to predict disease likelihood prior to the onset of clinical symptoms would allow for the implementation of preventative measures or early treatments, which are typically more effective than those initiated after disease manifestation. Next, we explore the PRS results for RDs in greater detail.

4.1. Rheumatoid arthritis

The heritability of RA family and twin studies has been estimated to be approximately 60% [55]. In addition, disease progression, clinical outcomes and specific RA phenotypes are also associated with genetic factors [56]. Predominantly, this genetic influence is attributed to the HLA region, which alone accounts for approximately one-third of the genetic predisposition [55]. The influence of the HLA region is notably marked by the shared epitope (SE) of the *HLA-DRB1* gene [57], which encodes a common amino acid sequence motif associated with RA susceptibility and progression, especially *HLA-DRB1*01* and *HLA-DRB1*04*. One restriction of SE is that it is primarily found in seropositive RA patients (particularly *anti*-CCP-positive patients); when autoantibodies are negative, heritability decreases [58,59]. Furthermore, the strong relationship between SE and *anti*-CCP means that autoantibody testing is preferred over genetic testing due to clinical feasibility [59–61].

Beyond the HLA region, the pathogenesis of RA involves multiple non-HLA genes that contribute to its complexity. Early GWAS identified significant non-HLA loci, such as the genes encoding peptidyl arginine deiminase 4 (*PADI4*), protein tyrosine phosphatase nonreceptor type 22 (*PTPN22*), and cytotoxic T-lymphocyte-associated protein 4 (*CTLA4*) [62,63]. These initial findings have been robustly confirmed in diverse cohorts [64]. Since 2007, the number of recognized non-HLA loci associated with RA has expanded significantly, with more than 100 identified through GWASs and subsequent meta-analyses [56]. In the most recent GWAS, which included a total of 276,020 participants across various ancestries from 37 cohorts (35,871 RA patients and 240,149 control individuals), the authors identified 124 independent non-HLA SNPs [65].

Ishigaki K et al. developed a PRS model based on multi-ancestry GWAS with a higher predictive performance than the models using only European data, providing new perspectives into RA aetiology and improving our ability to predict its genetic risk [65]. However, it showed limited effectiveness in predicting RA in African, South Asian, and Arab populations, reflecting the challenges in genetic prediction across diverse genetic backgrounds. The multi-ancestry PRS showed that the area under the curve (AUC) of the receiver operating characteristic (ROC) curve ranged from 0.59 to 0.66. Incorporating all known demographic, lifestyle, and genetic factors into the model increased the AUC to 0.77 [66].

Evaluating the contribution of independent SNPs for RA accounted for only 5% of the heritability that has been previously reported. This gap, known as “missing heritability,” is not exclusive to RA but is a recognized challenge across numerous health conditions, which hampers efforts to predict, prevent, and effectively treat these diseases [67].

Previously, Honda et al. explored the association between PRS and radiographic progression in RA patients [68]. Their findings revealed an association between PRS and severe radiographic progression, particularly in patients with younger-onset RA. The PRS effectively distinguished between patients with severe and nonsevere disease progression, underscoring its potential as a predictive tool in clinical settings. Although this study lacks validation in an independent cohort, it is pioneering in showing that an RA-PRS, constructed with disease susceptibility alleles and nonsignificant GWAS loci (43,784 SNPs in total), is associated with radiographic progression in RA patients.

Recently, Maurits et al. examined a PRS for differentiating between RA patients, clinically suspected arthralgia (CSA) patients and healthy controls [69]. Their PRS discriminated RA from CSA in both *anti*-CCP-positive and *anti*-CCP-negative strata, with a more explicit relationship in ACPA-positive patients. The same occurred with HLA-SE differences, which were more pronounced in the *anti*-CCP-positive strata and absent in the *anti*-CCP-negative strata. Furthermore, the scope of PRS research has expanded to explore genetic overlaps between RA and other conditions, including psychiatric disorders, osteoporosis, and other autoimmune diseases [70–72]. There are also studies that have explored the interplay between RA-PRS and environmental factors (air pollutants) [73].

4.2. Spondyloarthropathies

AS is a chronic inflammatory disease of unknown aetiology and the prototype of inflammatory pathologies of the musculoskeletal system affecting the axial skeleton, which are encompassed under the term axSpA. There is evidence supporting a significant genetic component involved in the development of AS [74]. AS occurs more frequently in families with affected members. Indeed, the risk (risk

ratio) of the disease in a sibling or first-degree relative of a patient has been estimated to be 80 for AS and 40 for SpA as a whole [75, 76]. Furthermore, AS has one of the highest heritabilities among complex phenotypes. AS is a complex disease in which multiple factors, both environmental and random, interact in genetically predisposed individuals, triggering pathological pathways leading to the development of the disease [77].

AS is one of the best examples of a disease associated with an HLA marker, the *HLA-B*27* allele [78]. The importance of evaluating the presence of the *HLA-B*27* allele in AS has already been discussed, although we still do not know the mechanisms by which it participates in the pathogenesis of the disease [79]. Other *HLA-B* alleles, as well as some class II HLA alleles, have been associated with susceptibility to AS [78]. This variability appears to be linked to the presence or absence of certain clinical characteristics [77]. The first GWAS including AS patients identified significant loci associated with this disease in the *ERAP1* and *IL23R* genes, marking the beginning of our understanding of the genetic foundations of the disease [80]. To date, researchers have identified more than 100 non-HLA loci associated with AS [81–84], revealing a complex genetic landscape that extends beyond the HLA region and significantly influences the development of this condition.

The first example of a PRS association with SpA was described by Rostami et al. [85], who demonstrated that a PRS based on the inclusion of 110 susceptibility SNPs, along with *HLA-B27* typing, showed a greater capacity to predict the risk of SpA than *HLA-B27* alone. By 2020, Knevel et al. [86] developed a tool (G-PROB) indicating that genetic data can discriminate between different diseases associated with similar signs and symptoms, specifically inflammatory diseases (RA, SLE, spondyloarthritis, psoriatic arthritis, and gout). This tool was validated by Hum et al. [87], who tested G-PROB in an independent population. They demonstrated that G-PROB effectively aids in the diagnosis of patients with early inflammatory arthritis, emphasizing its utility in a clinical setting.

Recently, Li et al. defined a PRS, validated in European and Asian populations, that showed a greater discriminatory capacity for AS than C-reactive protein, magnetic resonance imaging of the sacroiliac joints, or *HLA-B27* [88]. This finding suggests a promising role for this PRS in early diagnosis and in facilitating tailored therapeutic interventions, potentially transforming clinical practices in rheumatology. This PRS could be utilized to screen younger individuals experiencing chronic back pain before the age of 45, identifying those at increased risk for the disease for referral to specialized care, and aiding in the diagnosis of condition. A subsequent study by the same authors highlighted the clinical potential of this PRS in distinguishing different forms of axSpA [89]. Their findings demonstrate that PRS could effectively differentiate male patients with nr-axSpA from healthy first-degree relatives with notable accuracy. However, the effectiveness of PRS in making similar distinctions among female patients was less pronounced. These results underscore the importance of considering genetic factors, sex differences, and radiographic variations in clinical assessments and illustrate how a PRS could enhance diagnostic and treatment strategies in axSpA patients.

Acute anterior uveitis (AAU) is an intraocular inflammatory disease and a common comorbidity of disease within the SpA group, particularly AS. AAU manifests in 30%–50% of individuals with AS, and both conditions are strongly associated with *HLA-B27*, suggesting a common aetiological basis [90]. Huang XF et al. reported the first GWAS for AAU, confirming a strong overlap in its etiopathogenesis with that of AS and identifying new susceptibility loci for AAU [91]. They developed a PRS based on GWAS data that demonstrated high accuracy in identifying individuals at elevated risk for either AS with AAU or AS alone.

PsA is a heterogeneous, chronic inflammatory disease mediated by the immune system and characterized by inflammation of joints and entheses accompanied by dactylitis and spondylitis. PsA is also associated with nonmusculoskeletal comorbidities, including cardiovascular diseases, inflammatory bowel disease, depression, and anxiety [92]. Early diagnosis of PsA and the subsequent early initiation of therapy are crucial for preventing side effects and unnecessary pain or any potential irreversible joint damage [93].

The biological mechanisms underlying the development of PsA are incompletely understood; however, compelling evidence from family studies supports a significant genetic component [94]. The strongest genetic signals in PsA come from the HLA region, with significant associations found with specific alleles such as *HLA-C*06:02*, *HLA-C*08:01*, *HLA-B*27:05*, *HLA-B*38:01*, and *HLA-B*39:01* [95]. These alleles have been associated not only to disease susceptibility but also to distinct clinical features of PsA, underscoring the heterogeneity within HLA associations [96]. While further research is required to clarify the role of HLA alleles in determining the disease phenotype, there appears to be great heterogeneity in how HLA alleles are associated with the development of PsA.

The evidence also indicates the involvement of non-HLA genes in the pathogenesis of this disease. The initial GWAS in PsA identified crucial roles for HLA genes and confirmed associations with genes such as the interleukin 23 receptor (*IL23R*) and interleukin 12B (*IL12B*) [97]. Subsequent GWASs and meta-analyses have expanded these findings, identifying approximately 30 susceptibility non-HLA genes associated with PsA [95,98]. Recently, Soomro et al. carried out the largest GWAS to date [99], including 5065 PsA cases and 21,286 controls, alongside a comparison of 4340 PsA cases against 6431 patients with cutaneous-only psoriasis. The authors identified the NF- κ B and WNT signalling pathways as key pathways that differentiate PsA from cutaneous-only psoriasis, but only one novel susceptibility locus for PsA risk was described. Predicting the risk of PsA in patients with psoriasis continues to be a critical area of research. Patrick MT et al. reported that a genetic risk score based on 200 markers had the ability to distinguish between PsA and cutaneous-only psoriasis [100]. They demonstrated that genetic differences between the two phenotypes were due to regulatory elements, evidencing the need to integrate other omics data (transcriptomics and epigenomics inter alia), to make the implementation of personalized medicine a reality. Conducting additional GWASs exclusively with PsA patients is essential, not only for identifying genes but also for identifying PRSs. Future research should aim to integrate clinical data with genetic insights and incorporate advanced omics technologies to discover new biomarkers and improve prediction models.

4.3. Other rheumatic diseases

SLE is a chronic autoimmune systemic disease that predominantly affects women and has a wide range of clinical manifestations, affecting the skin, joints, and kidney. A delay in diagnosis or inadequate treatment may predict a poor outcome and even death. As in

other autoimmune diseases, the pathogenesis of SLE depends on genetic, epigenetic, and environmental factors, such as the regulation of the immune response and the influence of specific hormone levels [101].

As in other autoimmune diseases, HLA genes are involved in the pathogenesis of SLE. Specifically, HLA-DR2 and HLA-DR3 are clearly associated with SLE susceptibility, and HLA-DR is strongly associated with the presence of *anti-Ro/La* antibodies [102]. Efforts have been made to clarify the role of genetics in SLE pathogenesis through the development of several GWASs. Hom G et al. reported the first GWAS for SLE, analysing 5018 individuals of European origin, 1435 patients with SLE and 3583 control subjects [103]. These authors confirmed previously associated loci such as *HLA-DR*, *STAT4*, and *IRF5* but also identified the association of novel loci such as *C8orf13-BLK* and *ITGAM-ITGAX*. More recently, Wang YF et al. carried out a GWAS in which they included 11,283 SLE patients and 24,086 controls from China and described 38 additional loci associated with susceptibility [104]. The authors showed that their PRS yielded significantly improved results when trained on ancestry-matched datasets.

In a recent study developed by Khunsriraksakul C et al., a multi-ancestry and multi-trait meta-analysis of GWASs was performed by assembling multi-ancestry SLE GWAS datasets from European, Eastern Asian, and admixed American ancestries with a total of 21,625 SLE patients and 696,841 controls [105]. The authors identified 106 significant loci, 16 of which were considered novel. This comprehensive study allowed them to develop a risk model that showed consistent improvement in prediction accuracy when the PRS was added to the prediction model. Specifically, the AUC of using PRS + ANA + *anti-dsDNA* was 0.75, which improved the AUC of using only ANA + *anti-dsDNA* (0.73).

SSc pathogenesis is characterized by the progressive development of fibrosis in connective tissue involving internal organs. SSc, like many other immune-mediated inflammatory diseases, is the result of complex interactions between the environment and many additive genetic factors [106]. For SSc, two different types of disease have been clearly identified. Both are characterized by the presence of specific autoantibodies [107]. When *anti-topoisomerase* antibodies (ATAs) are present, patients show diffuse effects on the skin (dcSSc), clinically significant pulmonary fibrosis (PF) and a poor prognosis. Anti-centromere antibodies are associated with limited cutaneous systemic sclerosis (lcSSc), with a limited effect on the lung and a better prognosis.

Different studies have sought to estimate the heritability of SSc and have consistently found that, while lower than in other autoimmune diseases, the risk significantly increases in siblings of patients [108,109]; for example, Arnett et al. reported a 15-fold increased risk in siblings of SSc patients [110]. However, given the very low prevalence of the disease in the general population (0.0026%), the absolute risk in relatives remains relatively low (1.5–1.7%) [111]. The strongest genetic association observed is with the HLA genes [112]. Specifically, *HLA-DPB1*13:01* has been associated with ATA, while *HLA-DQB1*05:01* and *HLA-DQB1* alleles encoding a nonleucine residue at position 26 (*DQB1*26 epi*) have been associated with ACA. Additionally, *anti-RNA polymerase III* (ARA) antibodies have been associated with *HLA-DRB1*04:04*, *HLA-DRB1*11*, and *HLA-DQB1*03* alleles [113]. Although diverse GWASs have significantly enhanced our understanding of SSc pathogenesis, early studies' discovery rate was constrained by the small sample sizes of the study cohorts [111]. However, more recent studies have overcome this limitation.

López-Isac et al. carried out a GWAS and meta-analysis of previous GWASs, including 9095 SSc patients and 17,584 healthy controls [114]. They reported 27 signals independently associated with SSc, revealing molecular pathways potentially involved in vasculopathy and fibrosis, two hallmarks of the disease. A subsequent study performed by Bossini-Castillo et al. developed a PRS that showed its efficacy in discriminating patients with SSc from healthy controls and patients with other immune-mediated inflammatory diseases (IMiDs), such as RA and Sjögren's syndrome [115]. The PRS model incorporating 33 SNPs showed reasonable predictive accuracy (AUC = 0.67), but when it was combined with demographic and immunological parameters, the model's performance improved, achieving an AUC = 0.79, suggesting its potential to support the early and differential diagnosis of SSc. Recently, Ishikawa et al. reported significant findings from the largest Asian GWAS on SSc to date, which included 1428 cases and 112,599 controls [116]. They identified 6 novel susceptibility loci relevant to the disease. Moreover, the analysis demonstrated that their PRS could effectively predict disease susceptibility when the authors combined European and Japanese effect sizes (AUC = 0.60), revealing a shared genetic architecture of SSc between European and Japanese populations.

5. Challenges and future directions

PRSs are demonstrating substantial promise across a spectrum of RDs. While significant attention has been given to RA, SpA, LES and SSc in this chapter, PRSs also have potential in conditions where no score has yet been reported or where the approaches are still very preliminary [117]. However, there are universal challenges that must be addressed across different diseases.

One of the main challenges in applying PRSs in rheumatology is the significant scarcity of genetic data from diverse populations [118]. Most GWASs, and consequently PRSs, have predominantly involved participants of European descent. This lack of diversity presents a major obstacle to the universal applicability of PRSs across global populations [119]. Consequently, there is a need for more inclusive genetic studies that encompass underrepresented populations. Additionally, the current genetic models frequently fail to account for gene–environment interactions, which are crucial for understanding multifactorial diseases in which different genes may play crucial roles depending on the environmental risk factors and geographical location. It is crucial not to assume that a high genetic risk guarantees the manifestation of the phenotype. Depending on the disease being studied, environmental factors may be more important or even more significant than genetic risk factors. Therefore, expanding the GWAS to include more ethnically diverse populations and improving the statistical methodologies to integrate environmental and lifestyle factors into prediction models are essential steps towards addressing these challenges [45].

Another significant challenge in the clinical application of PRS is the “missing heritability,” where the genetic risk factors identified through current methodologies account for only a portion of the genetic predisposition to diseases [67]. The benefits and limitations of GWAS for identifying new SNPs linked to disease phenotypes are well established [120]. Increasing the sample size in the GWAS is

likely to facilitate the identification of additional loci associated with the different RDs and can be incorporated into PRS calculations. Significantly, some conditions are now being studied with sample sizes exceeding 1 million participants [121]. To achieve such large cohorts, international collaboration is essential. Moreover, structural variants such as copy number variants, deletions, and inversions have been poorly investigated [122] and could uncover additional layers of genetic risk.

AS exemplifies the potential clinical use of PRS in rheumatology. Current biomarkers for AS have been reviewed in this chapter and are moderately effective in diagnosing the disease and they are also used in prognostication and predicting treatment responses [74]. Given the heterogeneity of axSpA, there is a clear need for more informative biomarkers. Early treatment can significantly improve long-term patient outcomes [123,124], emphasizing the need for improved diagnostic and detection methods. This is particularly relevant given the significant diagnostic delays in AS, which range from 7 to 11 years in most countries and have not decreased over time despite extensive basic and clinical research efforts [125,126]. There is also a need to develop a strategy for biomarker discovery for use in detecting healthy populations or patients with early chronic back pain to diagnose or predict the development of AS.

The strong association of HLA alleles with susceptibility to RD, particularly between HLA-B*27 and AS, supported the introduction of the HLA-B*27 test in the 1970s of the 20th century, 50 years ago [21]. Since then, genetics tests have been included as usual tools for rheumatic disease diagnosis, beginning with AS and including others afterwards. However, as an increasing number of associations of non-HLA loci with RD have been identified and it has been demonstrated that their inclusion in GWASs increases their capacity for discrimination, the necessity of new, more complex tests has emerged [54]. A significant advantage of PRSs is their capacity to match or surpass the predictive power of biomarkers and other indicators currently used in clinical risk models across various diseases [127]. This capability indicates that PRSs have the potential to significantly enhance existing disease risk prediction models. PRS demonstrates greater predictive power in diseases with low prevalence and high heritability, such as most rheumatic diseases, and specifically AS [128]. For example, the widely used CRP test for distinguishing inflammation from other causes of back pain has an AUC of 0.7 for diagnosing AS [129]. Most PRSs for rheumatic diseases exhibit greater discriminatory capacity than this [54]. The key difference between a genetic test and other biomarkers is that genetics does not depend on the presence or activity of the disease to provide information. However, a major challenge for implementing PRS in a clinical context is that most PRSs have been developed within research study cohorts. The potential health benefits of PRSs will not be realized without a more comprehensive evaluation that includes an appropriate clinical context. We need to measure their utility in clinical practice by combining them with existing tests. Additionally, the use of other omics methods in parallel with this validation may facilitate the identification of new biomarkers [74].

In summary, PRSs demonstrate high discriminatory capacity for various RDs, with existing examples of near evidence of clinical utility, as in the case of AS. However, further basic and clinical studies are required to define appropriate genetic risk prediction models and the specific contexts for their use. Also, external validation in different populations is needed to assess the applicability of a specific PRS model in diverse mixtures of ancestries. As increasingly robust GWASs are conducted, future PRSs will enable more precise risk stratification by incorporating familial, environmental and lifestyle risk factors. This integration will conform to a global risk score, enhancing the ability to predict individual risk. The utility of PRS in rheumatology extends beyond diagnosis, encompassing the prediction of disease development, stratification of at-risk populations, and optimization of screening protocols.

6. Practice points

- Current biomarkers show considerable limitations in specificity and predictive value, leading to challenges in accurately diagnosing rheumatic diseases.
- Delays in the diagnosis of rheumatic diseases and in the establishment of adequate treatment cause a deterioration of joints and other tissues that is typically irreversible.
- Polygenic risk scores are valuable for predicting the genetic predisposition to rheumatic diseases, potentially allowing for earlier and more precise diagnoses. However, their integration into clinical practice requires further validation.

7. Research agenda

- Further research is needed to improve the accuracy and applicability of polygenic risk scores across different ethnic groups, as most current data are derived from populations of European descent.
- Research into how environmental and lifestyle factors interact with genetic predispositions to affect disease risk could lead to more comprehensive risk models.
- There is a need for longitudinal studies that track patients over time to validate the predictive power of polygenic risk scores in the real-world onset and progression of rheumatic diseases. Similarly, evaluating the cost-effectiveness of implementing polygenic risk scores in clinical practice will be relevant.

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& editing, Writing – original draft, Supervision, Conceptualization. **Roberto Díaz-Peña:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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